
Should This Concept Enter Memory?

Placebo-Calibrated Predictive Compression at Inference Time

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Abstract

When a user teaches a language model a novel term, a persistent-memory system must decide whether the model inferred the intended meaning before saving it. Existing write gates score confidence, consistency, novelty, or source trust; each can approve a confident, self-consistent misreading. We instead ask whether a candidate meaning compresses the user’s subsequent usage. For each candidate, we measure held-out usage log-loss reduction and standardize it against six surface-matched placebo meanings. We introduce gavagai pairs: teaching sets on which the intended meaning and a tempting misreading are extensionally identical, followed by usage that separates them. In a confirmatory test fixed before model evaluation (90 primary pairs, five unseen generator seeds), placebo-calibrated gain distinguishes the intended meaning from the misreading with AUC 0.949 (95% hierarchical bootstrap CI [0.916, 0.977]) on Qwen2.5-1.5B and 0.909 [0.860, 0.955] on Qwen2.5-3B, and 0.811 [0.740, 0.878] on the held-out SmoLM2 family. A ConsistencyGate-style support score obtains 0.370, 0.440, and 0.581; semantic entropy obtains 0.230, 0.249, and 0.439. A threshold selected only on development data transfers to unseen items with 55.6% true-concept admission at 9.5% all-negative false admission on the 1.5B model, but requires model-specific recalibration at 3B. In an end-to-end stress test, writing the first generated candidate hurts held-out log loss by 2.28 nats on average, whereas admitted candidates improve it by 1.44 nats with 92% precision at 27.8% coverage. Crucially, six observations do not repay a full textual description code on any test item; our result supports predictive compression as an admission signal, not a claim of immediate two-part MDL payback. Code, frozen splits, and every failed condition are included.

1 Introduction

Suppose a user says that a yellow wooden bowl is dax, and later that a yellow metal cup is also dax. Did dax mean yellow, wooden, bowl, or some conjunction? A model can be confident in the wrong answer because the teaching examples do not distinguish these hypotheses. If a persistent-memory system writes the first plausible interpretation, the error outlives the conversation and may become harder to correct.

We study the decision immediately before such a write. The model has inferred one or more candidate meanings from a teaching interaction and then observes a small buffer of the same interlocutor’s subsequent uses. No answer key says what the word means. The gate must accept a candidate, reject it, or wait for more evidence. The desired signal is therefore not whether the model already endorses the candidate, but whether the candidate predicts the user’s behavior better than reasonable alternatives.

Prediction and compression give a simple operational test: the reduction in log loss after conditioning on a candidate is exactly the number of nats it saves when coding the observed usage (Delétang et al., 2024; Cover & Thomas, 1999). Yet raw predictive gain is poorly

calibrated across items: some definitions are easy for the base model to follow, while others barely change its predictions. We standardize each candidate against six automatically constructed placebo meanings. The resulting score, z_{pc} , asks whether the candidate compresses usage unusually well for this item, rather than whether its raw gain exceeds a global scale-dependent number.

Evaluating this question requires ambiguity by construction. Ordinary concept-learning benchmarks usually reveal the intended label in their examples; then a confident-support score is often enough. Our gavagai pairs instead provide a true meaning M and a tempting misreading M' that make identical predictions on every teaching example. Only later uses separate them. Ground truth is available to the evaluator but never to the gate. This creates the precise failure mode that a persistent write policy must survive: a hypothesis can be novel, simple, highly supported, and wrong.

Our contributions are:

1. a controlled benchmark of five ambiguity families, with a frozen confirmatory split across five unseen generator seeds and explicit scope diagnostics;
2. a placebo-calibrated predictive-compression score and development-only admission protocol, evaluated against direct support, entropy, surprise, simplicity, and raw-gain baselines; and
3. an end-to-end stress test that extracts usage decisions from natural utterances, samples candidate definitions, and applies the frozen gate before evaluation.

We also report three negative results that materially narrow the claim. A threshold does not transfer unchanged across model scales; relational meanings are outside the reliable scope of the tested base models; and the predictive savings from six observations never repay a full textual description.

2 Related work

Persistent memory and write gates. Context distillation, adapters, and trained prefixes can turn contextual information into reusable state (Snell et al., 2022; Chen et al., 2024; Eyuboglu et al., 2025). The orthogonal question is what should be admitted. Continual-learning and agent-memory systems gate writes using randomness or representativeness, geometric novelty, normalized loss, Bayesian surprise, and source trust (Prabhu et al., 2020; Aljundi et al., 2019; Wang et al., 2026; Li, 2025; Gorlo et al., 2026; Zahn & Chana, 2026; Yang et al., 2026). These signals answer whether an item is useful, new, or trustworthy, not whether a candidate interpretation predicts the teacher’s later usage.

The closest locus is ConsistencyGate (Zhang & Li, 2026), which places a ground-truth-free gate before an inference-time persistent write. Consensus gates also filter self-training data (Huang et al., 2022; Stein et al., 2026). Our novelty claim is therefore only about the criterion: held-out predictive compression relative to item-matched placebos. We implement the latency-oriented support probability of ConsistencyGate as a direct baseline.

Uncertainty and compression. Semantic entropy detects inconsistent generations (Farquhar et al., 2024), but an over-specific misreading can be both wrong and low-entropy. Agreement is likewise not a standalone correctness certificate (Ding, 2026). Rate-distortion views of agent memory ask what to retain under a budget or how to preserve the agent’s own decisions (Colaco & Lahjouji, 2026; Zou et al., 2026). We instead score a candidate before it is written and use another party’s later word choices as evidence. We test both predictive savings and explicit two-part codes, and keep their conclusions separate.

3 Method

3.1 Predictive compression from later usage

For one interlocutor and novel word, let T be teaching examples and let c be a candidate meaning induced from T . After induction, the system buffers n later usage events $D = \{(x_j, y_j)\}_{j=1}^n$, where $y_j \in \{0, 1\}$ indicates whether the interlocutor applies the word to x_j . Let $q_0(y | x, T)$ be the base prediction and $q_c(y | x, T, c)$ the prediction when conditioned on c . The predictive compression gain is

$$\Delta(c; D) = \text{NLL}_0(D) - \text{NLL}_c(D) = \sum_{j=1}^n \log \frac{q_c(y_j | x_j, T, c)}{q_0(y_j | x_j, T)}. \quad (1)$$

Positive gain means the candidate shortens a prequential code for observed usage. The buffer is temporally later than induction but earlier than persistent consolidation; deploying the gate thus trades immediate writing for evidence.

3.2 Item-matched placebo calibration

Raw gain depends on prompt sensitivity and the model’s ability to follow a definition. For each item we automatically generate $K = 6$ placebo candidates: two wrong atomic meanings, one wrong conjunction, two over-specific meanings, and one over-broad meaning. They match the surface form and complexity range of plausible hypotheses without inspecting the true meaning. Let μ_i and s_i be the sample mean and standard deviation of their gains. Our primary score is

$$z_{\text{pc}}(c) = \frac{\Delta(c) - \mu_i}{s_i}. \quad (2)$$

On development data only, we select a single acceptance threshold that maximizes true-candidate admission subject to false admission of M' and placebos not exceeding 10%. The threshold is then frozen. This is a selective policy: low coverage is acceptable if admitted candidates are reliable.

Not all compression scores are two-part MDL. Equation 1 is predictive code savings, and Equation 2 is a calibrated signal; neither subtracts the cost of specifying c . We separately evaluate $G_L(c) = \Delta(c) - L(c)$ using (i) an index into the shared candidate pool, (ii) the reference model’s NLL for the textual statement, and (iii) a fixed-width token code. We call a concept “paid for” only when the corresponding $G_L > 0$.

3.3 Baselines

All scores see the same candidate pairs. Support is the model probability that a candidate is consistent with all teaching examples, matching the single-pass log-probability form of ConsistencyGate. Semantic entropy is negative mean Bernoulli entropy of conditioned held-out usage predictions. Surprise is mean Bernoulli KL divergence from the no-definition predictions. Simplicity is negative reference-model description length. We also report raw gain Δ and the best-placebo margin. Appendix A gives exact prompts.

4 Experimental design

4.1 Gavagai pairs

Each item contains a pseudoword, six teaching examples, an intended meaning M , a misreading M' , six placebo meanings, and three to six held-out uses. By construction, M and M' label every teaching example identically and disagree on held-out examples. The primary domain contains: (G1) an atomic property versus a conjunction; (G2) a superordinate category versus the single observed kind; and (G3) a material versus a perfectly correlated object kind. G4 tests argument order and G5 absolute versus relative size. Pilot experiments, conducted before the confirmatory split, showed unreliable definition following on

Table 1: Schematic G1 gavagai pair. The gate observes the teaching set and later usage, but never the evaluator-only names M and M' .

stage	described object	Tovi says dax?	M : yellow	M' : yellow $\wedge$ wooden
teaching	yellow wooden bowl	yes	yes	yes
teaching	yellow wooden cup	yes	yes	yes
teaching	red metal bowl	no	no	no
later	yellow metal cup	yes	yes	no
later	red wooden cup	no	no	no

G4 and prior override on G5; we therefore declared G1–G3 primary and retained G4–G5 as scope diagnostics.

Table 1 illustrates why ordinary validation on teaching examples is insufficient. Both candidates achieve perfect teaching accuracy, and the conjunction may receive higher support because every positive happens to be wooden. The later yellow metal object supplies the information that the teaching set lacks. Placebos are generated from fixed templates (for example, red, wooden, yellow and shiny, and a thing) rather than chosen after scoring.

The development set is the existing seed-11 set (40 items, eight per family). The frozen test uses five unseen seeds {101, 211, 307, 401, 503}, six items per family and seed: 150 items total, of which 90 are primary. Test examples were not inspected when selecting thresholds. We evaluate canonical and dictionary-style definition prompts. The frozen models are Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, and SmolLM2-1.7B-Instruct; all are used without parameter updates. The 1.5B Qwen model alone selects thresholds.

4.2 Statistics and contamination control

The primary comparison is M versus M' on unseen G1–G3 items. We report pooled AUC, paired wins, a one-sided sign test fixed before evaluation, and 95% hierarchical bootstrap intervals that resample generator seeds and then items within a seed. Every prompt, model, seed, and failed family is retained. All-negative false-admission includes M' and all placebos, preventing a relative ranking metric from hiding a gate that always accepts something. Before teaching, the model also scores held-out labels using each pseudoword alone; near-chance AUC and tokenization statistics test for accidental pretraining associations.

4.3 End-to-end interface stress test

P12 supplies candidate meanings and structured usage labels to isolate the admission criterion. P13 adds two interfaces. First, each usage decision is rendered as one of four natural utterance templates; a probability threshold selected on development utterances extracts whether the speaker applies the word. Second, the 1.5B model samples four concise candidate definitions from teaching examples only. The gate selects the candidate with largest z_{pc} and accepts it only at P12’s frozen threshold. Evaluation uses true labels only afterward. The frozen primary generation seed is reported separately from two explicitly post-hoc decoding-seed checks. This remains a templated microworld stress test, not a natural-conversation benchmark.

5 Results

5.1 Predictive compression separates intended meanings

On 90 unseen primary pairs, z_{pc} reaches AUC 0.949 on Qwen2.5-1.5B and 0.909 on 3B (Table 2); paired wins are 87/90 ($p = 9.82 \times 10^{-23}$) and 80/90 ($p = 5.26 \times 10^{-15}$). Performance is not driven by one generator seed: 1.5B canonical AUC is 0.926–0.975 across the five seeds, with 17 or 18 wins out of 18 in each. A dictionary-style prompt yields 0.968 and 0.893, so the signal is not tied to the canonical wording. On the held-out SmolLM2 family, canonical AUC is 0.811 ([0.740, 0.878], 74/90 wins, $p = 2.19 \times 10^{-10}$); changing only to the predeclared

Table 2: Unseen G1–G3 test, canonical prompt. Intervals hierarchically resample generator seeds and items. Support is the direct ConsistencyGate-style baseline. SmolLM results are filled by the same frozen protocol and never used for calibration.

Model	placebo-z	raw gain	support	sem. entropy	surprise
Qwen2.5-1.5B	.949 [.916,.977]	.874	.370	.230	.526
Qwen2.5-3B	.909 [.860,.955]	.851	.440	.249	.732
SmolLM2-1.7B	.811 [.740,.878]	.662	.581	.439	.580

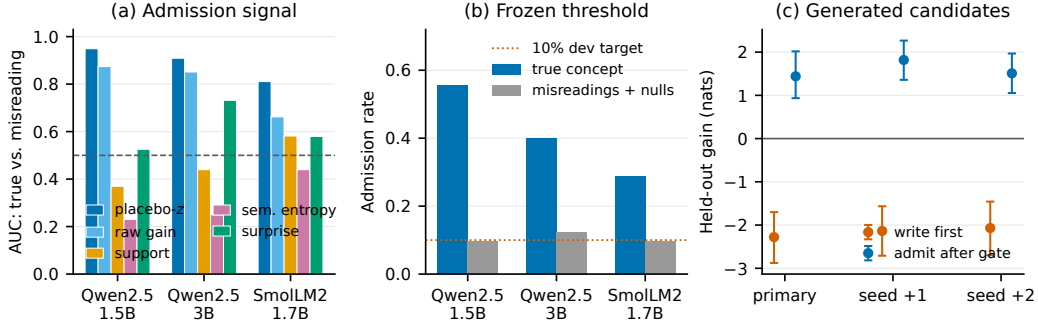


Figure 1: (a) Direct baselines on unseen candidate pairs. Dashed line is chance. (b) One admission threshold selected on 1.5B development data and applied without retuning. (c) End-to-end generated candidates: first-candidate write-all versus candidates admitted by the frozen gate; bars are 95% item-bootstrap intervals. The first seed is confirmatory; the other two are post-hoc robustness.

dictionary prompt improves it to 0.912 [0.828,0.973]. The cross-family effect is therefore positive but more prompt-sensitive.

The direct baselines clarify the mechanism. Support is below chance on both Qwen scales because the narrower misreading is fully consistent with teaching and often easier to endorse. Semantic entropy is more strongly inverted: confidence rewards the over-specific interpretation. Surprise improves at 3B but remains below calibrated gain. Simplicity reaches 0.751 at 1.5B (Appendix Table 3); it captures part of the atomic-versus-conjunctive structure but cannot resolve the material-kind family. Figure 1 summarizes these comparisons.

Placebo calibration, rather than predictive loss alone, accounts for a substantial part of the improvement: at 1.5B, raw gain obtains 0.874, subtracting the best placebo obtains 0.943, and standardizing by the placebo distribution obtains 0.949. The placebos measure whether a particular prompt/model pair can express the hypothesized distinction on that item. This also explains the hard boundary on G4 below: when all definitions fail to control predictions, normalization cannot create missing conditional behavior.

5.2 A development-frozen threshold works within scale, not across scales

The development rule chooses $z_{pc} > 1.143$ (development TPR 50.0%, FPR 9.5%). On unseen 1.5B canonical items it admits 55.6% of true meanings, 3.3% of misreadings, and 10.6% of placebos; the combined all-negative FPR is 9.5%. Under the dictionary prompt, the corresponding rates are 47.8% and 8.9%. Thus the frozen threshold transfers across items and prompt wording at the development model scale.

At 3B, true admission falls to 40.0% and all-negative FPR rises to 12.4%. More importantly, a development-selected absolute “decidability” threshold retains 67.8% of 1.5B pairs at 98.4% pair accuracy but retains none at 3B. A dimensionless score does not guarantee cross-model calibration: deployment requires calibration data for each base model. We therefore retract the earlier cross-scale threshold claim and treat AUC, not a universal threshold, as the cross-model result. On SmolLM2, the same canonical threshold preserves

the intended error rate (9.7% all-negative FPR) but admits only 28.9% of true meanings; the dictionary prompt yields 57.8% TPR and 8.9% FPR. Thus a conservative threshold can transfer accidentally well in error rate while its operating coverage remains model- and prompt-dependent.

5.3 Predictive savings do not repay a textual definition in six uses

Subtracting the shared candidate-index cost yields positive net gain for 47.8% of true 1.5B candidates and 65.6% at 3B. In contrast, neither the reference-model text code nor the fixed-width text code is repaid by any true candidate. For items with positive predictive gain, a stationary-use extrapolation gives a median text-code payback horizon of 84.3 observations (IQR 60.3–132.7) at 1.5B, 38.6 (19.6–105.4) at 3B, and 339.5 (187.9–558.4) on SmolLM2. This extrapolation is not observed future performance. The supported claim is that predictive gain identifies the better hypothesis; whether it is economical to store a full natural-language definition depends on representation cost and expected future use.

5.4 Generated candidates turn write-all loss into selective gain

Natural-utterance extraction obtains AUC 0.99996 and 99.38% accuracy on 480 unseen utterances, using a threshold chosen on 128 development utterances. This near-ceiling result reflects explicit affirmative/rejection templates and should not be read as open-domain semantic parsing.

The candidate generator is substantially less reliable, which makes the admission decision meaningful. In the frozen run, writing its first definition decreases held-out compression gain by 2.278 nats on average (95% item-bootstrap CI $[-2.877, -1.701]$). The frozen gate admits 25 of 90 items (27.8% coverage); 23 of 25 admitted candidates are beneficial (92.0% precision), and their mean gain is +1.441 nats $[+0.935, +2.017]$. Across two post-hoc decoding seeds, coverage is 32.2–35.6%, precision 87.5–93.1%, write-all gain -2.136 to -2.069 , and admitted gain +1.510 to +1.818. Selection regret versus the best of four candidates is at most 0.001 nats on average because extracted labels are almost exact; candidate quality, not selection among candidates, is the current bottleneck.

5.5 Scope and contamination

The declared G4 argument-order diagnostic is at chance (1.5B AUC 0.559, 95% CI $[0.454, 0.673]$), showing that compression cannot rescue a model that does not condition on the candidate definition. G5 reaches 0.849 but is excluded from the primary claim because pilots revealed prior override. Before teaching, the pseudowords alone do not separate primary labels: per-family AUC is 0.510–0.533 at 1.5B and 0.454–0.531 at 3B. Only 3.3% of pseudowords are single tokens (mean 2.2 tokens per word). On SmolLM2, primary-family zero-shot AUC is 0.500–0.508 and only 0.7% are single tokens. Its G5 zero-shot AUC is 0.946, however, and its post-definition G5 candidate AUC is inverted at 0.261; both were retained as declared diagnostics and further support excluding G5 from the primary claim. These controls make memorized word meanings an implausible explanation, while not proving absence from all pretraining data.

Error analysis. All three 1.5B pairwise errors occur in G2 category-level ambiguities: two stick versus cane items and one toy versus kite item. Their score gaps are small (0.01–0.35), but one has high base decidability, showing that uncertainty alone would not catch every semantic error. The two harmful P13 admissions are also G2. Both generated descriptions are over-specific (a kite-like object and a neck-worn textile), and both lie near the frozen threshold ($z = 1.57$ and 1.20). More diverse negative placebos or a minimum absolute-gain condition are promising, but tuning them on these errors would invalidate the confirmation; we leave them as future hypotheses.

6 Limitations and implications

The study uses small instruction-tuned models, English prompts, a finite symbolic microworld, and binary usage decisions. Its strongest evidence is for choosing between supplied hypotheses; P13’s generated candidates remain templated and low-coverage. Natural conversation may contain indirect, pragmatic, drifting, or strategically misleading usage. The method also needs later observations, so it cannot validate a concept immediately after a single explanation.

Predictive fit is not truth. The gate estimates an interlocutor-specific convention, including a stable false belief; source trust and safety policies remain separate layers. Repeated evaluation on the same buffer may overfit candidate search, requiring a fresh validation suffix or sequential testing in longer interactions. A real memory system must additionally enforce per-user isolation, deletion, privacy, and correction. Appendix D summarizes our storage experiments; they show why admission and representation should be evaluated separately.

Despite these limits, the negative results point to a practical design. Buffer evidence; compare multiple induced hypotheses using future usage; calibrate against matched placebos; abstain by default; encode frequently reused concepts compactly; and recalibrate whenever the base model or prompting interface changes.

7 Conclusion

A confident interpretation is not necessarily one worth writing. On controlled ambiguities, placebo-calibrated predictive compression distinguishes intended meanings from self-consistent misreadings where support and entropy fail, and it converts harmful write-all candidate generation into a selective positive-gain policy. The result is narrower than immediate MDL consolidation: full textual descriptions do not repay their code cost in six uses, and thresholds are model specific. That separation—predictive evidence first, storage economics second—is the central lesson for inference-time concept memory.

Reproducibility statement. The repository includes the frozen protocol timestamped before confirmatory evaluation, deterministic data generators and exact splits, all prompts, raw per-item probabilities and scores, development-only threshold selection, hierarchical-bootstrap code, environment versions, and negative conditions. No API or proprietary model is required. Appendix E lists commands and artifact paths.

LLM usage disclosure. An OpenAI Codex model assisted substantially with experiment implementation and orchestration, statistical analysis code, figure generation, literature organization, and manuscript drafting. The human authors directed the research questions and are responsible for independently inspecting the code, verifying the reported artifacts, validating citations and claims, and approving the submitted manuscript.

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A Exact prompts and scores

For candidate c , the canonical definition prefix is “In Tovi’s usage, [word] means [meaning].” The dictionary variant is “Tovi’s dictionary entry: [word] — [meaning].” Each observation then asks whether Tovi would apply the word to the described object and scores the next-token probabilities of Yes and No, renormalized over those two tokens. The base condition states only that the word is unfamiliar. Support receives the teaching examples and candidate statement and asks whether the meaning is consistent with every example. P13 induction asks for one concise dictionary-style meaning consistent with all teaching examples and samples four sequences at temperature 0.8, top- p 0.9, capped at 18 new tokens.

Table 3: All primary pair scores, Qwen2.5 canonical prompt. AUC compares M against M' .

Score	Qwen 1.5B		Qwen 3B		SmolLM 1.7B	
	AUC	wins	AUC	wins	AUC	wins
placebo-z	.949	87/90	.909	80/90	.811	74/90
raw gain	.874	87/90	.851	80/90	.662	74/90
best-placebo margin	.943	87/90	.810	80/90	.769	74/90
model-text net gain	.918	89/90	.882	86/90	.758	76/90
support	.370	35/88	.440	43/89	.581	43/76
semantic entropy	.230	18/90	.249	13/90	.439	29/90
surprise	.526	51/90	.732	68/90	.580	57/90
simplicity	.751	74/90	.779	80/90	.741	74/90

B Frozen protocol and additional results

The confirmatory protocol was written before any P12 model run. Development uses generator seed 11. Test uses seeds 101, 211, 307, 401, and 503. The direction of the sign test, primary G1–G3 domain, two prompt variants, three models, all-negative FPR, code-length definitions, and interpretation rules were fixed in advance. The rules explicitly require removal of cross-family or two-part-MDL claims if the corresponding tests fail.

Table 4: Frozen-threshold unseen-test admission. The threshold is selected only on Qwen2.5-1.5B canonical development data. “Neg.” pools M' and six placebos.

Model	prompt	M TPR	M' FPR	placebo FPR	all-neg. FPR
Qwen2.5-1.5B	canonical	.556	.033	.106	.095
Qwen2.5-1.5B	dictionary	.478	.033	.098	.089
Qwen2.5-3B	canonical	.400	.000	.144	.124
Qwen2.5-3B	dictionary	.456	.011	.146	.127
SmolLM2-1.7B	canonical	.289	.056	.104	.097
SmolLM2-1.7B	dictionary	.578	.022	.100	.089

Table 5: Canonical placebo-z by primary ambiguity family on Qwen2.5-1.5B.

Family	n	AUC [95% hierarchical CI]	paired wins
G1 atomic vs. conjunction	30	.950 [.874,1.000]	30/30
G2 category vs. kind	30	.937 [.849,.984]	27/30
G3 material vs. kind	30	1.000 [1.000,1.000]	30/30

C Admission-signal taxonomy

Table 6: Representative write or consolidation signals. Our distinction is the criterion, not the existence or inference-time locus of a gate.

Family	signal	typical locus
GDumb / reservoir	randomness or class balance	replay write
GSS / MIR	gradient diversity or interference	replay write/retrieval
SAGE	geometric novelty	online write
Self-Sizing Hopfield	normalized-loss hysteresis	online write
Worth Remembering	Bayesian surprise	online write
Write-Time Gating / TRUSTMEM	source trust or faithfulness	agent-memory write
GATES / LMSI	sampled self-consistency	training-data admission
ConsistencyGate	candidate support probability	inference-time persistent write
DeMem	rate-distortion of agent decisions	budgeted compaction
Ours	held-out user-usage compression vs. placebos	inference-time persistent write

D Storage substrate boundary

Separate pilot experiments (P3-P11) tested whether context-derived behavior could be stored in a small per-user parameter delta. Deltas persisted and were isolated by user identifier, but a word-substitution control showed that the strongest delta stored a word-independent bias over objects rather than a lexical binding: replacing the taught pseudoword with an opposite-meaning word barely changed its effect. Token-triggered lexical entries restored binding but lost part of the cross-format transfer of context injection. We therefore do not present the substrate as concept learning and do not use it in the admission results. The lesson is methodological: persistence, partition isolation, lexical binding, and transfer need separate controls.

E Reproduction

From research/learning-in-referencing, create the pinned environment in requirements.txt. Then run `p12/generate_splits.py`, one invocation of `p12/eval_gate.py -model MODEL` per listed model, `p12/contamination.py -model MODEL`, and `p12/analyze.py`. Run `p13/run_endtoend.py` for the frozen seed and listed robustness seeds, followed by `p13/analyze.py`. Figure data are regenerated by `paper/figures/make_iclr_figures.py`. Raw JSON retains every item, candidate, score, model, prompt variant, split, and generator seed.